



Original Article

Dose and dose rate dependence of the tissue sparing effect at ultra-high dose rate studied for proton and electron beams using the zebrafish embryo model

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ABSTRACT

Purpose: A better characterization of the dependence of the tissue sparing effect at ultra-high dose rate (UHDR) on physical beam parameters (dose, dose rate, radiation quality) would be helpful towards a mechanistic understanding of the FLASH effect and for its broader clinical translation. To address this, a comprehensive study on the normal tissue sparing at UHDR using the zebrafish embryo (ZFE) model was conducted.

Methods: One-day-old ZFE were irradiated over a wide dose range (15–95 Gy) in three different beams (proton entrance channel, proton spread out Bragg peak and 30 MeV electrons) at UHDR and reference dose rate. After irradiation the ZFE were incubated for 4 days and then analyzed for four different biological endpoints (pericardial edema, curved spine, embryo length and eye diameter).

Results: Dose-effect curves were obtained and a sparing effect at UHDR was observed for all three beams. It was demonstrated that proton relative biological effectiveness and UHDR sparing are both relevant to predict the resulting dose response. Dose dependent FLASH modifying factors (FMF) for ZFE were found to be compatible with rodent data from the literature. It was found that the UHDR sparing effect saturates at doses above ~ 50 Gy with an FMF of ~ 0.7–0.8. A strong dose rate dependence of the tissue sparing effect in ZFE was observed. The magnitude of the maximum sparing effect was comparable for all studied biological endpoints.

Conclusion: The ZFE model was shown to be a suitable pre-clinical high-throughput model for radiobiological studies on FLASH radiotherapy, providing results comparable to rodent models. This underlines the relevance of ZFE studies for FLASH radiotherapy research.

Introduction

The FLASH effect, describing the sparing of normal tissue while maintaining the tumor killing efficiency at ultra-high dose rate (UHDR), has received great attention in the radiotherapy community and has

stimulated research in all disciplines involved. This particular interest came because of its potential to widen the therapeutic window: the normal tissue complication probability (NTCP) curve is shifted to higher doses while the tumor control probability (TCP) is unaffected [1]. Although first patients are treated with UHDR regimes [2,3], it would be

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preferable to understand the underlying mechanisms in order to know the optimal beam parameters for triggering the FLASH effect [4,5] prior to a broader clinical translation. Another important aspect to be understood is why the sparing effect is not observed in tumors [5].

In recent years several pre-clinical studies investigated the tissue sparing by performing biological experiments with different doses [6] and dose rates [7–9]. Furthermore, several experiments [10–14] were conducted to study if the ionization density, typically quantified by the dose-averaged linear energy transfer in water (LET_d), has any effect on the magnitude of the FLASH effect (as on most radiobiological effects). However, the role of the radiation quality is not conclusively understood up to now. UHDR experiments at very high LET_d using carbon ions also showed a sparing of normal tissue [15,16], but did not yet allow a quantitative comparison with low LET beams.

Most of the studies that have reported a tissue sparing effect at UHDR have used rodent models. However, rodent experiments typically suffer from small animal numbers due to the large efforts associated with animal care and ethics regulations that limit the number of animals to a minimum. Therefore, many of the previous rodent FLASH studies were restricted to a small dose range or even to single dose points [6]. *In vitro* experiments using e.g. cell cultures [17] would allow a higher throughput, but they are probably not sufficient to fully understand the mechanisms behind the FLASH effect because it is considered to be an *in vivo* effect that cannot be reflected fully by individual cells but needs a tissue and organ structure [4].

Therefore, several recent FLASH studies made use of the zebrafish embryo (ZFE) model [8,9,13,18–22]. It is an *in vivo* model, which is not subject to the same regulations as rodent experiments and less labor intensive than conventional *in vivo* models based on rodents in terms of handling and follow up. The radiation response of ZFEs is considered to be qualitatively comparable to mammals which has motivated its use in radiotherapy research, e.g. for the screening of radioprotectors or sensitizers [23]. The large number of eggs laid by zebrafish and their small size allow radiobiological experiments with high throughput [24]. Therefore, the ZFE model can reduce the need for rodent experiments and at the same time help to obtain statistically significant results at manageable effort for radiobiological questions where it is a suitable surrogate. Furthermore, the ZFE model can be used to generate hypotheses which can then be further studied in rodent experiments, with the limitation that the short 5-day follow up allows to assess only acute radiation effects.

This work presents a systematic experimental study towards a better understanding of the normal tissue sparing at UHDR in dependence of physical beam and irradiation parameters (radiation quality, dose, dose rate) using the ZFE model, in continuation of previous studies on the topic [9,18,20,21]. One-day-old ZFE were irradiated with protons at the University Proton Therapy Dresden (UPTD) or with electrons at the ELBE (Electron Beam of high Brilliance and low Emittance [25]) research accelerator at Helmholtz-Zentrum Dresden-Rossendorf (HZDR) and their development and morphology were analyzed 4 days after irradiation. The magnitude of normal tissue sparing could be studied by analyzing how *NTCP* curves for different ZFE endpoints were modified when irradiated at UHDR compared to a low dose rate reference. Different radiation qualities and a wide span of dose rates were available in the different irradiation setups: a monoenergetic 225 MeV proton beam, a proton spread out Bragg peak (SOBP) and monoenergetic 30 MeV electrons. Using the ZFE model, the radiation response to single dose exposures at UHDR and reference dose rate was studied over a dose range of 15–95 Gy for four different biological endpoints. It is shown that ZFE irradiated at UHDR exhibited a lower toxicity compared to those irradiated at reference dose rate, whereas the magnitude of the sparing effect varied strongly among the three beams studied. The comparison of the two different proton beam qualities revealed that proton relative biological effectiveness (*RBE*) and UHDR sparing are competing effects and that their combination determines the resulting radiation response. Altogether, the UHDR sparing data obtained in this

work for ZFE is in good quantitative agreement with rodent data from the literature.

Materials and methods

Zebrafish embryo model

In this work, one day old ZFE were irradiated using methods already described [9,18,20,21,26]. ZFE irradiations were performed on consecutive days applying a timed procedure starting with transport of wildtype AB zebrafish embryos from the Center for Regenerative Therapies TU Dresden (CRTD) to the respective facility followed by maintenance at room temperature (23–24 °C). Approximately 1 h before irradiation, about 30 ZFEs were sealed in 0.5 ml Eppendorf tubes, each filled with 200 µl low melting agarose (UltraPure Agarose, Invitrogen, Germany) and 300 µl E3 embryo medium [27]. Such a group of 30 ZFEs is referred to as one “sample”. Irradiations were performed at reference dose rates and UHDR at oxygen levels in the surrounding medium of < 10 mmHg. The oxygenation was controlled daily using dedicated, unirradiated control samples and an OxyLite sensor (Oxford Optronix Ltd, Abingdon, UK). The irradiations were performed in up to 10 runs per day, which took about 6 h, i.e. with embryos of 24–30 h post-fertilization (hpf). The influence of embryo ageing during irradiation was minimized by alternating of reference and UHDR irradiations in each run. Two to three independent experiment replications on separate days were performed for each beam modality. After irradiation, the ZFE were transferred in 96 well plates and followed for 4 days under standard conditions (28 °C). At the end of follow up, the development of pericardial edema and curved spine were recorded by microscopic observation and a photo of each individual embryo was taken. From the microscopic images the length deficit and the eye diameter deficit of the 5 days-old embryos were assessed relative to unirradiated controls in a blinded study. Comparing the dose–effect curves for the different endpoints of ZFE irradiated at UHDR with those irradiated at reference dose rate provided information about the magnitude of the normal tissue sparing effect in different tissues and organ systems.

Irradiation setups, beam parameters and dosimetry

Irradiations of ZFE were performed at the fixed beamline in the research area of UPTD in two different setups (proton entrance channel and SOBP) using mean dose rates of 240 Gy/s (entrance channel) and 600 Gy/s (SOBP) and at the ELBE electron accelerator at HZDR using 0.9×10^5 Gy/s. These dose rates represent the maximum possible mean dose rates in the respective setups. For practical reasons, the ZFE were irradiated with doses of 15–50 Gy using electron beams and of 20–95 Gy with the proton beams. The radiation quality of the electron beam (LET_d : 0.27 keV/µm) and the proton entrance channel (LET_d : 0.4 keV/µm) are comparable while the SOBP beam has a considerably higher ionization density (LET_d : 2.5 keV/µm at the center of the sample). All irradiation parameters are summarized in Table 1.

Further technical details on the different experimental setups, accelerators and dosimetry methods can be found in references [9,18,20,28] and in supplement S1.

Data analysis

Each ZFE was analyzed concerning the four endpoints described above and for each sample the relative proportion of ZFEs with pericardial edema and/or a curved spine as well as the mean embryo length and the mean eye diameter were calculated for the surviving embryos at 4 dpi (days post irradiation). The mean embryo length and eye diameter were normalized to the average values of the unirradiated control ZFEs of the irradiation day at 5 dpf and subtracted from 100 % to obtain the relative deficit compared to the controls. Those average values of the different samples were then used as data points for the further analysis.

Table 1

Summary of irradiation parameters for ZFE experiments using three different beams.

	monoenergetic protons	proton SOBP	monoenergetic electrons
Energy	225 MeV	modulated from 225 MeV	30 MeV
Facility / accelerator	UPTD / isochronous proton cyclotron	UPTD / isochronous proton cyclotron	ELBE / superconducting electron linac
LET _d	0.4 keV/μm	2.5 keV/μm	0.27 keV/μm
Ultra-high dose rate			
Mean dose rate	240 Gy/s	600 Gy/s	0.9×10^5 Gy/s
Bunch dose rate	1.3×10^3 Gy/s	3×10^3 Gy/s	1.4×10^9 Gy/s
Reference dose rate			
Mean dose rate	0.33 Gy/s	0.5 Gy/s	0.1 Gy/s
Bunch dose rate	1.65 Gy/s	2.5 Gy/s	1,500 Gy/s

The data points as a function of dose D were fitted using a $NTCP$ model based on Poisson statistics [29,30] having the form $NTCP = a \cdot 2^{(-\exp(b \cdot (1-D/c)))}$ with the three free parameters a , b and c . For the pericardial edema and curved spine endpoints parameter a was fixed at 1 because they are classical $NTCP$ endpoints where very high doses lead to a 100 % probability of occurrence. For the length and eye diameter endpoints, which do not represent a probability for a certain radiation effect but describe the severity of a radiation damage (or developmental delay), the parameter a was not fixed but also fitted by the optimizer. Its numerical value reflects the maximum possible effect to the ZFE (maximum relative growth delay) that can be induced by the irradiation. The statistical analysis is described in detail in supplement S2.

Böhlen et al. recently published a database of FLASH *in vivo* data (mostly from experiments with rodents) and analyzed the magnitude of the sparing effect at UHDR as a function of dose [6]. In their publication, a quantity called FLASH modifying factor (FMF) was introduced which translates doses applied at UHDR to equivalent doses at conventional dose rate that cause the same effect (for a given biological system and endpoint), comparable to the concept of relative biological effectiveness [31]. To obtain a quantitative measure of the sparing effect, we derived such FMF values from the fitted ZFE dose–effect curves (as already demonstrated for a first example in a previous letter [21]) and compared them with the *in vivo* database compiled by Böhlen et al. [6]. The uncertainty of the obtained FMF values was estimated from the uncertainties of the fit functions.

Results and discussion

A compilation of the dose–effect curves measured for ZFEs for four different biological endpoints for three different beams (Table 1) is depicted in Fig. 1 at reference dose rate and UHDR. The shown plots contain data from about 9,000 ZFEs (each datapoint is the average of about 30 embryos). Embryonic survival at the time of analysis (4 dpi) was almost 100 % at doses up to 50 Gy and still about 80 % at 95 Gy (not shown).

For all three beams, a significant sparing effect at UHDR compared to the reference irradiations on certain biological endpoints can be observed (curved spine and eye diameter for the proton entrance channel, curved spine and embryo length for the SOBP and all four endpoints for electrons). However, the magnitude of the sparing effect is considerably different between the irradiations with the two proton beams and the electron beam. Whereas for the latter a significant sparing effect at UHDR was proven for all endpoints, for the proton irradiations it is significant only for two out of four endpoints and in general much less pronounced. For the curved spine endpoint, a significant UHDR sparing could be observed for all three beams. The much higher UHDR of the electron beams can be assumed to be one of the reasons for the considerably larger sparing effect compared to what was observed with the proton irradiations. As already shown in our previous study [9] and also by others [7,32], higher mean dose rates can amplify the UHDR sparing effect. The results obtained in the present experiments are in good agreement with our previous ZFE experiments [9,20,21] and also with experiments by others [8,22] using one day old ZFEs, but less well

comparable to those performed with ZFE irradiated at 4 hpf [13,19] exhibiting higher radiosensitivity.

The different endpoints are sensitive in different dose ranges: while the pericardial edema endpoint rises very steeply and is already saturated at 30 Gy, the embryo length endpoint requires about 60 Gy for reaching the plateau. FLASH studies using the ZFE model should analyze the endpoint that is best suited to the dose range used. The optimal endpoint to be used would depend on the dose range which should match with the rising edge of the $NTCP$ curve. Especially for studies with single dose points it is important to make sure that they lie on the rising edge of the dose–effect curve. While in this work, the curved spine and embryo length endpoints seem to be best suited, one of the other endpoints could be suitable as well if the dose points would be distributed differently.

The direct comparison of the curved spine and length deficit data at reference dose rate (Fig. 2a,b) shows that the dose–effect curves for monoenergetic protons (solid lines) and electrons (dotted lines) are overlapping while the curves for the SOBP irradiation (dashed lines) are shifted towards lower doses (highly significant ***) by about 10 %. This is likely due to the increased RBE of the SOBP irradiation ($LET_d = 2.5$ keV/μm) compared to the proton entrance channel ($LET_d = 0.4$ keV/μm) and high energy electron beam ($LET_d = 0.27$ keV/μm).

This result underpins the excellent dosimetric consistency between the different experimental setups and the different radiation qualities and shows that our ZFE model [9,18,20] is well under control, even though irradiations were performed in different laboratory environments at different accelerators [33].

It should be noted that the dose values for the proton irradiations are based on independent measurements with a PinPoint ionization chamber as well as EBT3 films, while the dosimetric reference for the electron irradiations were EBT3 films alone (see supplement S1) assuming that their response is independent of the dose rate. Different studies with both electron and proton beams confirmed the dose rate independence of the response of EBT3 films [34–36] and our own tests of EBT1 films at the ELBE accelerator [37] showed the same result. However, a recent experimental study for 70 MeV protons indicated a change of the response of EBT3 films by some percent at UHDR [38]. Even if the EBT3 films would show an over-response of 5–10 % in net optical density at UHDR (following reference [38]) the basic statements concerning the electron data of this work would remain valid.

The observed shift of the SOBP dose–response curves compared to the beams with lower LET demonstrates that the ZFE model can resolve differences in radiosensitivity (or tissue sparing) in the order of 10 %. For the other two studied endpoints (pericardial edema and eye diameter) that are not shown in Fig. 2, a higher effectiveness of the SOBP irradiation compared to the proton entrance channel can be noticed as well, but the results are not statistically significant. For the pericardial edema endpoint, this is probably because the dose points were not optimally distributed (see Fig. 1).

It is worth noting, that in our proton experiments the tissue sparing at UHDR partially or even entirely compensated the RBE increase in the SOBP compared to the entrance channel, as visible in Fig. 2c,d by comparing the dot-dashed (SOBP, UHDR) and solid (proton entrance

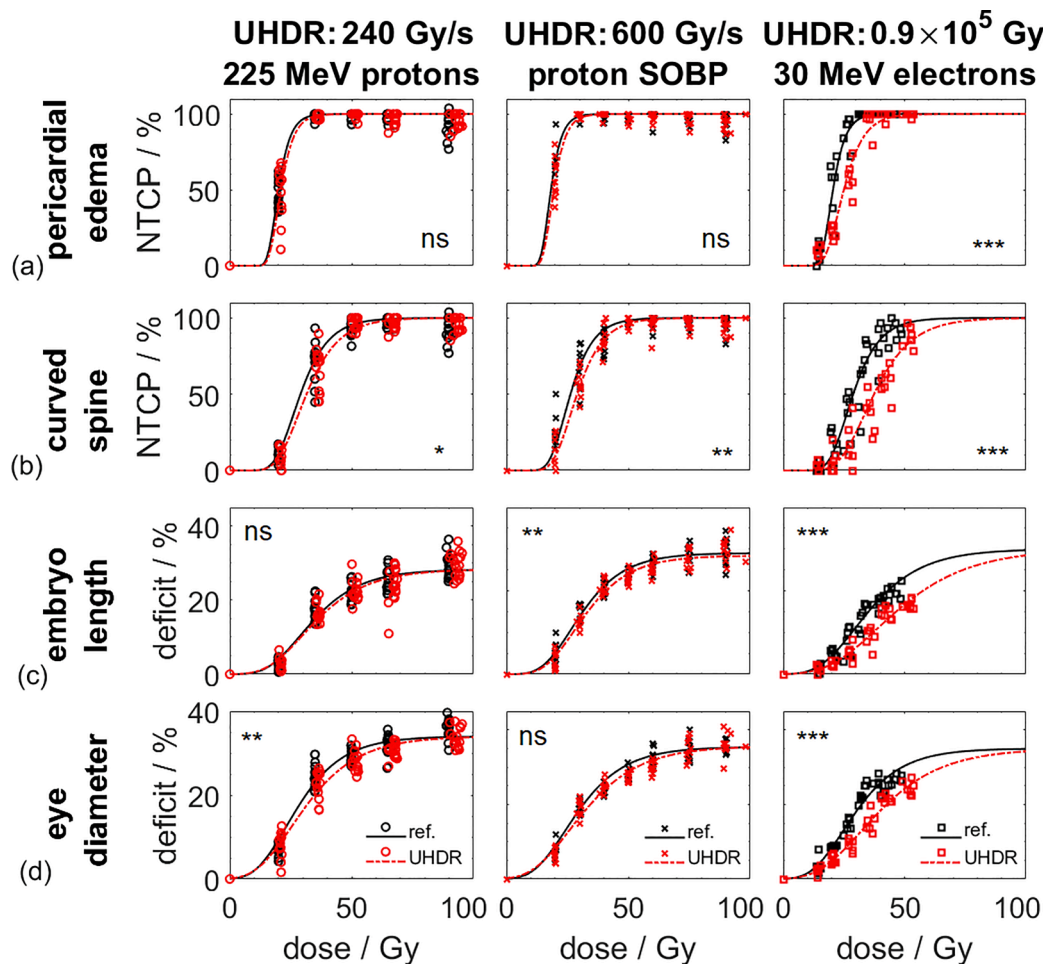


Fig. 1. Dose-effect curves determined for 5 days-old ZFE following reference (solid lines) and UHDR (dash-dotted lines) irradiation with 225 MeV protons (circles, left column), in a proton SOBP (crosses, middle column) and with 30 MeV electrons (squares, right column). The rows show the different endpoints: the fraction of ZFE with a pericardial edema (a), the fraction of ZFE with a curved spine (b), the deficit of embryo length compared to unirradiated controls (average length $\sim 3900 \mu\text{m}$) (c) and the deficit of eye diameter compared to unirradiated controls (average diameter $\sim 240 \mu\text{m}$) (d). The electron datapoints in panel (c) were already shown in a recent letter [21] as a preview on the full dataset presented here. The statistical significance of the sparing effect at UHDR is indicated by ns (non-significant), * (significant), ** (very significant) and *** (highly significant).

channel, reference dose rate) *NTCP* curves. Despite good evidence for an increasing *RBE* towards the end of the proton range from pre-clinical research and radiobiological considerations [31], it is currently not taken into account in dose prescription and treatment planning for proton therapy. Our results imply that it could be useful to also consider the variable proton *RBE* in addition to the UHDR sparing effect in proton FLASH treatment planning studies in order to understand the interplay of these two competing effects (*RBE* increases while FLASH decreases the biological effectiveness) and to predict the resulting radiation response correctly.

The FLASH modifying factor *FMF* as a function of dose for the irradiation with 30 MeV electrons at $0.9 \times 10^5 \text{ Gy/s}$ is shown in Fig. 3a for the four different ZFE endpoints considered in this work. For doses where the *NTCP* curves were already saturated for both UHDR and reference irradiations (depends on the endpoint, see Fig. 1), no *FMF* data points are shown because no reasonable tissue sparing information can be derived in this case. The ZFE proton data are not shown in this plot.

The ZFE *FMF* data are compared with rodent data derived by Böhlen et al. [6] from literature *NTCP* curves [1,32,39–50]. The ZFE data cover the entire dose range where rodent datapoints are available with a single experiment. The cloud of rodent data points contains *FMF* values for various acute and late endpoints (e.g. skin reaction, gut toxicity, survival, lung fibrosis, brain cognitive function), obtained from experiments where rodents were irradiated. The average dose rate was in most

of these experiments in the order of about 100 Gy/s ($\sim 40 - 340 \text{ Gy/s}$), with the exception of those by Ruan et al. [32] and Montay-Gruel et al. [44] where the irradiations were performed with dose rates exceeding 10^6 Gy/s . This *in vivo* literature database contains both electron and proton datapoints, which seem to overlap without a particular systematic. Therefore, the datasets for both beams are shown together and not distinguished in these plots. The rodent *FMF* data group around a bow-shaped curve that starts to become smaller than 1 (i.e. tissue is spared at UHDR) at doses exceeding $\sim 10 \text{ Gy}$ (typically considered as a loose threshold for the FLASH effect) and then decreases towards an *FMF* of $\sim 0.7-0.8$ at doses around 50 Gy .

The ZFE electron *FMF* values match very well with the rodent data from the literature (Fig. 3a) and follow the same trend. This confirms the suitability of the ZFE model for radiobiological studies investigating the FLASH effect and its capability to provide meaningful results for the application in rodents (and possibly also other mammals like humans). This can be very useful for future FLASH studies due to the advantages of the ZFE model (simple handling, high throughput, ethical aspects) compared to mouse experiments.

Compared to the experiments by other groups [13], where undifferentiated ZFE at 4 hpf were irradiated, the ZFE at 24 hpf used in the present work are in a developmental stage where the different organ systems are already differentiated [51]. Therefore, the ZFE experiments presented in this work allow the investigation of radiation effects on

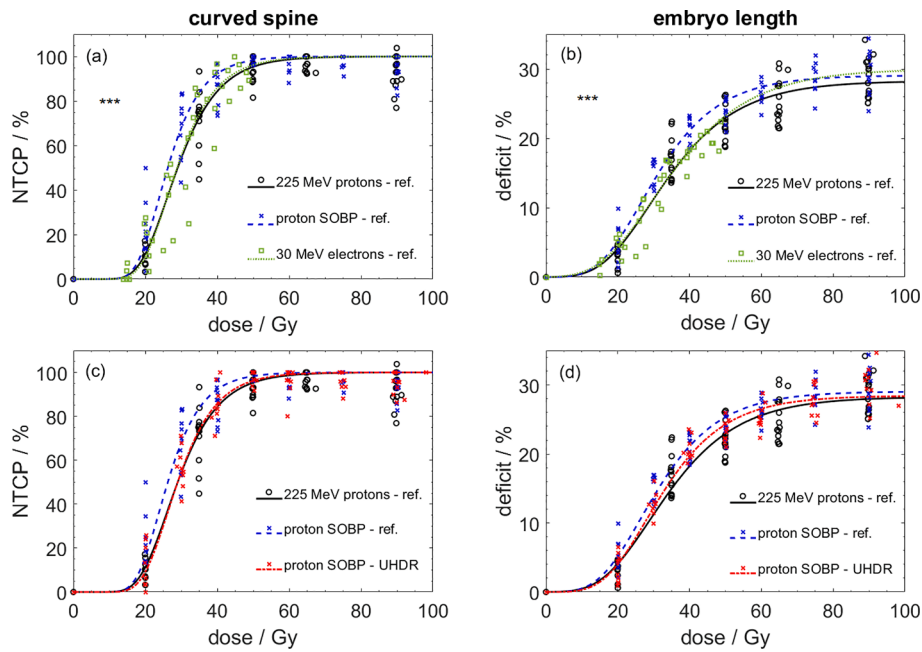


Fig. 2. Panel (a) and (b): Dose-effect curves determined for 5 days-old ZFE following irradiation with 225 MeV protons (solid lines and circles), in a proton SOBP (dashed lines and crosses) and 30 MeV electrons (dotted lines and squares) at reference dose rate (between 6 and 32 Gy/min). Panel (c) and (d): Dose-effect curves for irradiation at reference dose rate with 225 MeV protons (solid lines and circles) and in a proton SOBP (dashed lines and dark crosses) compared to the proton SOBP dose-effect curve at UHDR (dot-dashed lines and light crosses). Panel (a) and (c) show data for the curved spine endpoint and panel (b) and (d) for the embryo length deficit endpoint. The *** symbol indicates a highly significant RBE increase of the SOBP compared to the monoenergetic 225 MeV protons and 30 MeV electrons.

different organ systems via analysis of different endpoints. However, interestingly, the dose dependence of the sparing effect seems to be quite comparable for the four biological endpoints considered in the present study despite their different nature (cardiovascular, muscular-skeletal and developmental) and their different dose-effect characteristics (see Fig. 1). Differences between the endpoints are mostly visible at lower doses while at high doses the *FMF* curves as a function of dose converge.

As clearly visible in the ZFE electron data for all endpoints, the *FMF* as a function of dose follows a bowed curve shape, i.e. the tissue sparing effect at UHDR saturates at doses above ~ 50 Gy. This bowed curve can also be vaguely noticed in the rodent literature data but from the ZFE data it becomes really evident. The knowledge that the tissue sparing effect shows this remarkable dose dependence may be very useful towards a mechanistic understanding of the FLASH effect. For instance, a

recent radiochemical modeling study based on transient hypoxia [52] predicted a dose dependence that is qualitatively comparable to the trend shown in Fig. 3a.

Fig. 3b shows the curved spine *FMF* as a function of dose for the three different dose rates investigated (240 Gy/s, 600 Gy/s and 0.9×10^5 Gy/s). The curved spine was chosen for this comparison because it is the endpoint where a significant sparing could be observed for all three irradiation modalities (see Fig. 1). It can be observed that the tissue sparing effect is less pronounced for the two proton beams as compared to the electron irradiation at much higher dose rate. The sparing effect is more significant (smaller error bars) for the proton SOBP irradiation at 600 Gy/s (diamonds) than for the irradiation in the proton entrance channel at 240 Gy/s (circles), but only the 30 MeV electron irradiation at 0.9×10^5 Gy/s was able to trigger a strong UHDR sparing (squares). On

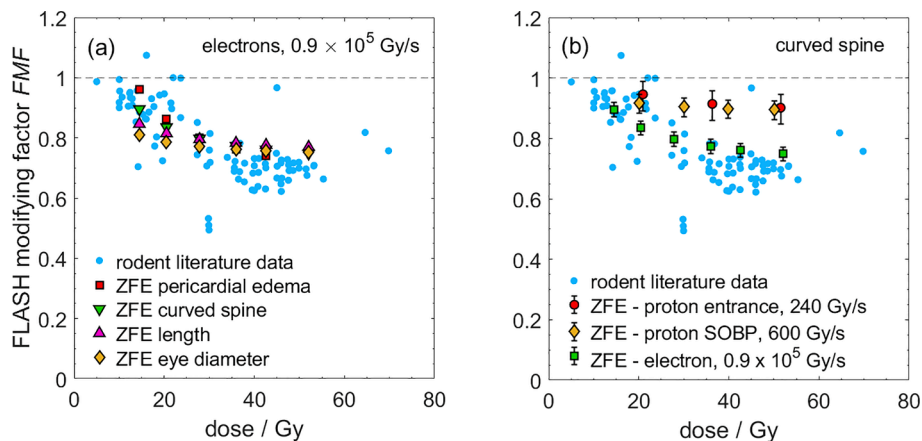


Fig. 3. Panel (a): FLASH modifying factor (*FMF*) as a function of dose obtained from the dose-effect curves of ZFE irradiated with 30 MeV electrons at UHDR and reference dose rate for four different endpoints: pericardial edema (squares), curved spine (inverted triangles), embryo length deficit (triangles) and eye diameter deficit (diamonds). Panel (b): *FMF* for the curved spine endpoint as a function of dose for ZFE irradiated with 225 MeV protons at 240 Gy/s (circles), in a proton SOBP at 600 Gy/s (diamonds) and with 30 MeV electrons at 0.9×10^5 Gy/s (squares). The ZFE data is compared with rodent *FMF* data derived by Böhlen et al. [6] from dose-effect curves found in the literature [1,32,39–50] (light dots). The error bars indicate the uncertainty of the *FMF* calculated from the dose-effect curves.

the other hand, as visible in the literature data (dots) the same magnitude of tissue sparing can be observed in rodents already at moderate UHDR in the order of 100 Gy/s. Most groups observed an UHDR tissue sparing for the ZFE model at mean dose rates in the range of kGy/s and higher [8,9,13]. Moreover, in particular for 1-day old ZFE, the irradiation at partial oxygen levels of $\leq 7\%$ in the surrounding medium intensifies the sparing effect of UHDR irradiation [8,20]. This partial oxygen level is comparable to human and rodent normal tissue [53].

The different dose rates required for ZFE and rodent experiments at comparable oxygenation level also points to a strong biological component in the mechanism of the FLASH sparing effect, besides processes on the radiochemical level that are usually suspected to be the main driver. A theoretical model of the FLASH effect must be able to explain these differences in UHDR required to trigger the tissue sparing in ZFE and rodents.

For both, electron and proton beams, it would be interesting to study more and comparable dose rates in the range between 600 Gy/s and 0.9×10^5 Gy/s in future experiments in order to better understand the dose rate dependence of the UHDR tissue sparing effect. This could be done for instance using electrons from the ELBE accelerator, where the dose rate can be tuned with high flexibility (from 0.1 Gy/s up to 10^5 Gy/s). Proton beams with dose rates noticeable above 600 Gy/s with a field size that is large enough for ZFE irradiations are not available at standard clinical proton therapy facilities like UPTD. It should however be noted that groups at other proton irradiation facilities were able to reach dose rates of a few kGy/s [8,36]. Another promising tool for radiobiological experiments at extremely high dose rates, even beyond the 0.9×10^5 Gy/s reached at the ELBE accelerator, would be laser-accelerated particles [33]. Despite the many drawbacks of laser-plasma accelerators (large energy and angular spread, pulse-to-pulse fluctuations, pulse repetition rate limitations) they can provide dose rates for radiobiological experiments that exceed all other accelerator technologies.

Conclusion

This work comprises a comprehensive study on the normal tissue sparing at UHDR as part of the FLASH effect using the zebrafish embryo model. The ZFE model has been proven to be suitable for radiobiological studies with high throughput. In the present experiments, four different biological endpoints were studied simultaneously with a single model covering a wide dose range. FLASH modifying factors obtained from these experiments are in good agreement with literature data for rodents, supporting the use of the ZFE model for pre-clinical research in the context of FLASH radiotherapy.

The results obtained in the present study give new insights into the tissue sparing at UHDR and therefore may contribute to a better understanding of the FLASH effect and its mechanisms. A pronounced dose rate dependence and a saturation of the tissue sparing at high doses were revealed. Moreover, it was shown that both proton RBE and UHDR sparing must be considered in order to accurately predict the resulting radiation response of a biological system to an UHDR proton beam.

The ZFE model can be an efficient and reliable tool also for future FLASH radiotherapy studies, e.g. to deepen the understanding of the observed dose rate dependence or to further explore the physical beam parameter space. The present work provides the basis for the design and planning of such experiments.

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CRediT authorship contribution statement

Felix Horst: Conceptualization, Formal analysis, Investigation,

Visualization, Writing – original draft. **Elisabeth Bodenstein:** Methodology, Writing – review & editing. **Michael Brand:** Methodology, Writing – review & editing. **Stefan Hans:** Methodology, Writing – review & editing. **Leonhard Karsch:** Investigation, Writing – review & editing. **Elisabeth Lessmann:** Methodology, Writing – review & editing. **Steffen Löck:** Formal analysis, Writing – review & editing. **Michael Schürer:** Methodology, Writing – review & editing. **Jörg Pawelke:** Conceptualization, Funding acquisition, Investigation, Supervision, Writing – review & editing. **Elke Beyreuther:** Conceptualization, Funding acquisition, Investigation, Project administration, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.radonc.2024.110197>.

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